Natural Gas & Cross-Commodity Volatility Regime Framework

Regime Classification, Shock Clustering & Cross-Asset Correlation Dynamics

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Executive Summary

This project develops a cross-commodity energy market framework designed to identify volatility regimes, systemic stress conditions, and shifting correlation structures across natural gas, crude oil, and power markets.

Using historical futures and proxy market data, the framework classifies natural gas volatility into distinct market regimes and evaluates how stress propagates throughout broader energy markets during periods of elevated uncertainty.

The analysis demonstrates several key findings:

- Natural gas volatility is regime-driven rather than constant
- Cross-asset shock events cluster during stress periods
- Correlation structures change materially between calm and stress environments
- Diversification benefits deteriorate during systemic volatility
- Energy markets exhibit asymmetric responses to macroeconomic and commodity-specific shocks

The framework combines volatility modeling, spread analysis, cross-commodity correlation analysis, and systemic shock detection into a unified institutional-style risk dashboard intended to replicate components of real-world energy trading and risk-monitoring workflows.

Market Structure Overview

The North American energy complex exhibits interconnected but structurally distinct behavior across crude oil, natural gas, and power markets.

Natural gas markets typically demonstrate the highest realized volatility due to weather sensitivity, storage dynamics, transportation constraints, and regional supply-demand imbalances.

Crude oil markets are more heavily influenced by macroeconomic conditions, geopolitical developments, and global demand expectations.

Power markets behave differently from traditional commodities due to the non-storable nature of electricity and the influence of grid constraints, weather patterns, and generation mix composition.

Because of these structural differences, volatility transmission across energy assets becomes especially important during periods of market stress.

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The initial stage of the framework classified natural gas volatility into distinct market regimes using rolling realized volatility measures and regime thresholding techniques.

The analysis demonstrates that volatility clusters into persistent stress periods followed by prolonged normalization phases rather than behaving as a stable or normally distributed process.

Figure 1 — Natural Gas Spark Spread Regime Framework

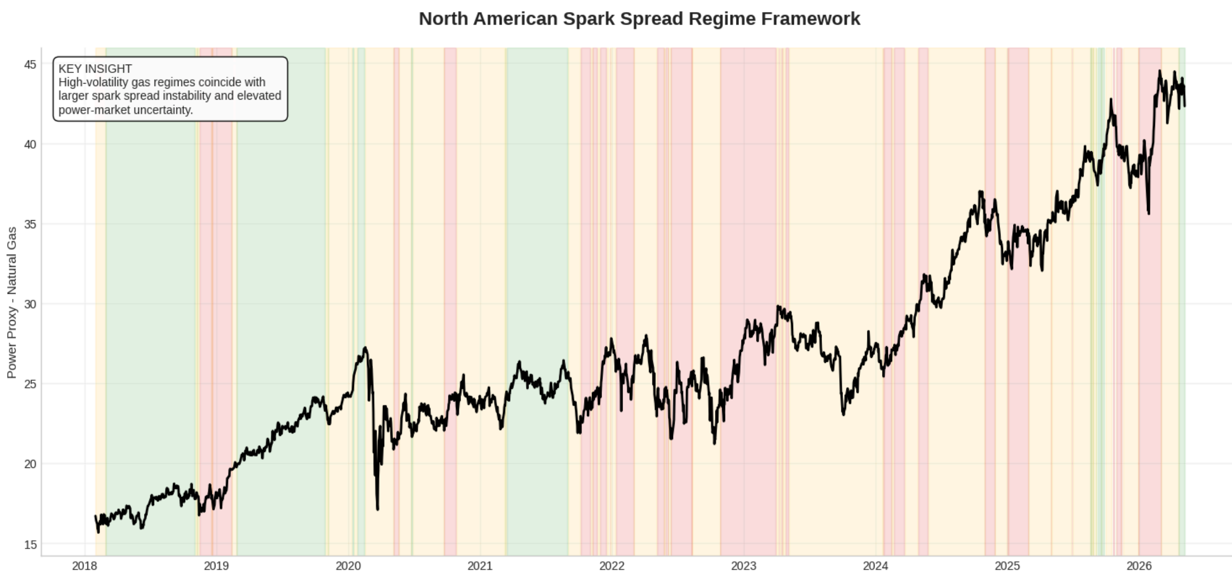

Figure Commentary

Periods of elevated natural gas volatility coincide with materially larger instability in spark spread behavior, suggesting tighter coupling between gas market stress and power-market uncertainty.

Stress regimes appear persistent rather than isolated, reinforcing the importance of regime-based market analysis over static volatility assumptions.

Cross-Commodity Energy System Analysis

To evaluate broader systemic behavior, the framework incorporated:

- WTI Crude Oil Futures
- Natural Gas Futures
- Utilities Sector ETF (XLU) as a power-market proxy

Each series was normalized to evaluate structural differences in trend behavior and volatility transmission across the energy complex.

Figure 2 — Cross-Commodity Energy Market Structure

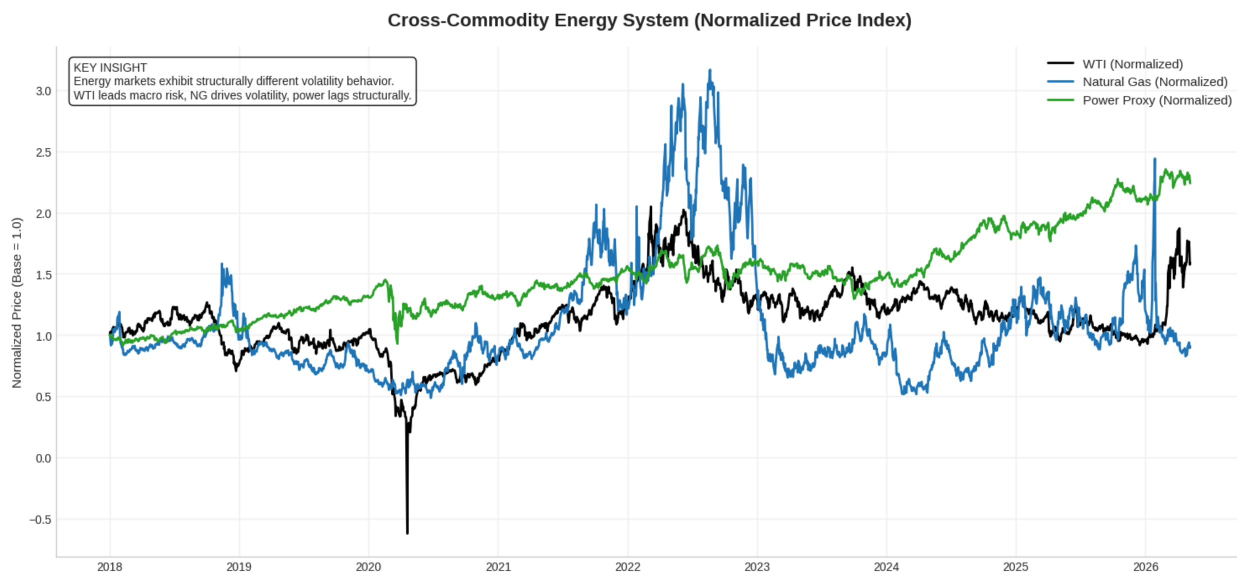

Figure Commentary

The normalized framework highlights the structurally different volatility profiles across crude oil, natural gas, and power markets.

WTI exhibits stronger macroeconomic trend behavior, while natural gas displays materially higher realized volatility and sharper stress transitions.

Power markets demonstrate lower realized volatility but remain sensitive to broader commodity-driven stress conditions.

Systemic Shock Detection & Risk Dashboard

The framework next evaluated simultaneous cross-asset shock behavior using rolling volatility-adjusted thresholds.

Shock events were defined as daily returns exceeding two standard deviations relative to rolling realized volatility estimates.

The analysis demonstrates that stress events frequently cluster across energy assets rather than occurring independently.

Figure 3 — Institutional Energy Risk Dashboard

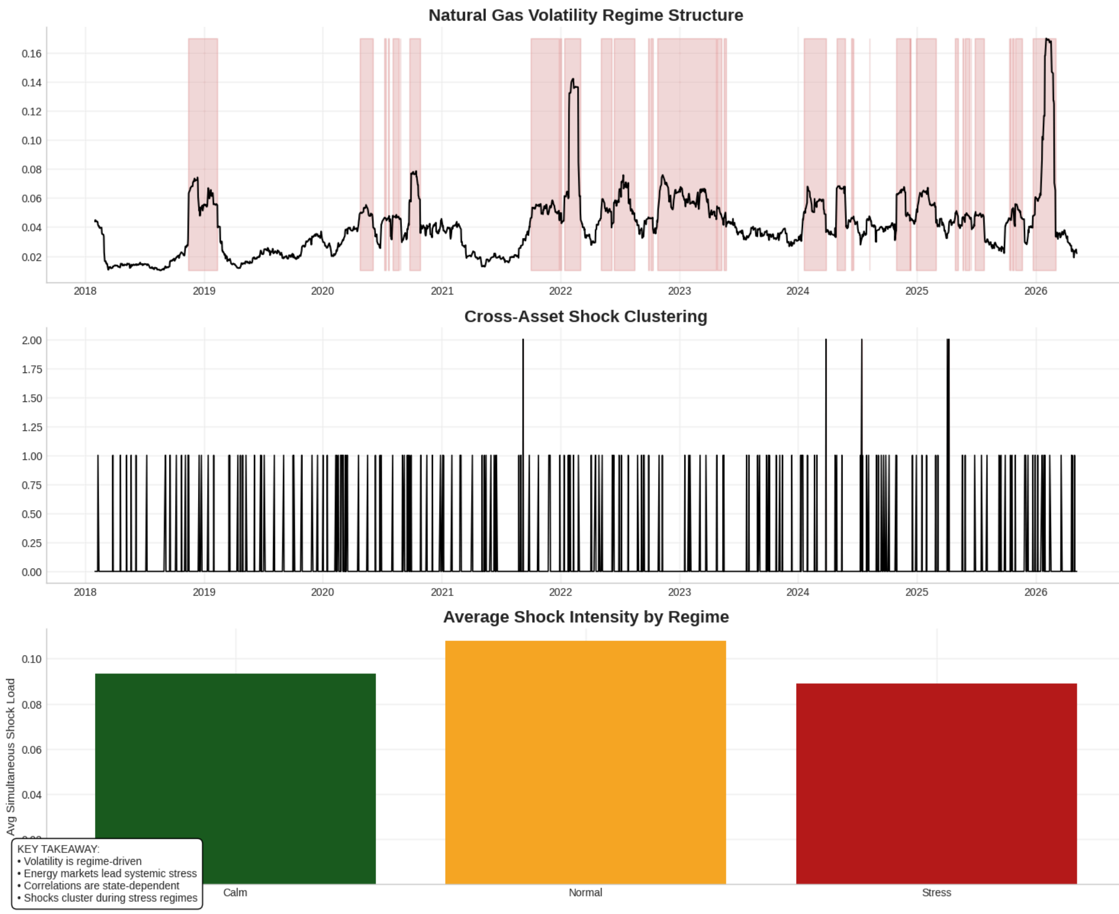

Figure Commentary

The dashboard framework highlights several important structural observations:

- Volatility regimes persist over time rather than appearing randomly
- Cross-asset stress conditions cluster during elevated volatility periods
- Systemic shock intensity rises materially during stress regimes
- Energy markets exhibit non-linear risk behavior during periods of instability

The incorporation of WTI volatility as a secondary confirmation layer improves broader macro stress identification.

Correlation Regime Analysis

One of the most important findings from the framework is that correlation structures are regime-dependent rather than static.

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Traditional diversification assumptions weaken materially during stress periods as cross-asset behavior converges.

Figure 4 — Correlation Structure by Market Regime

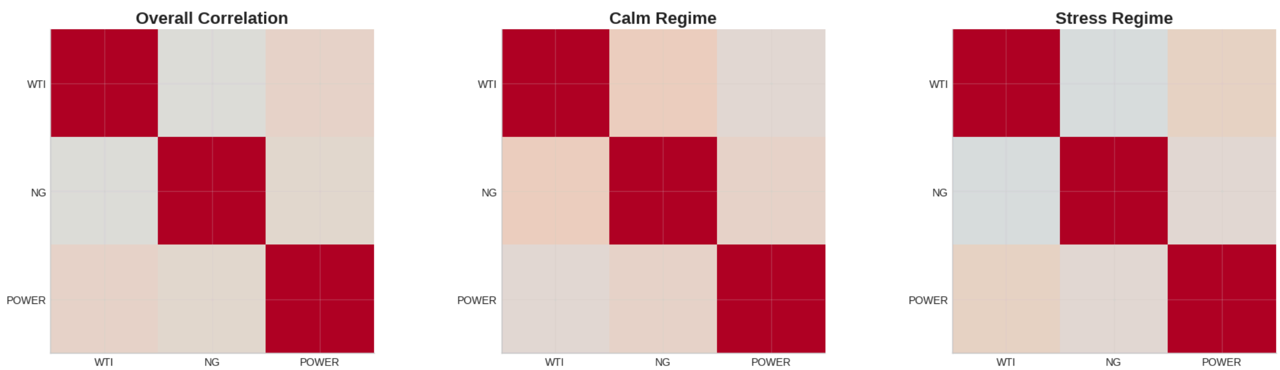
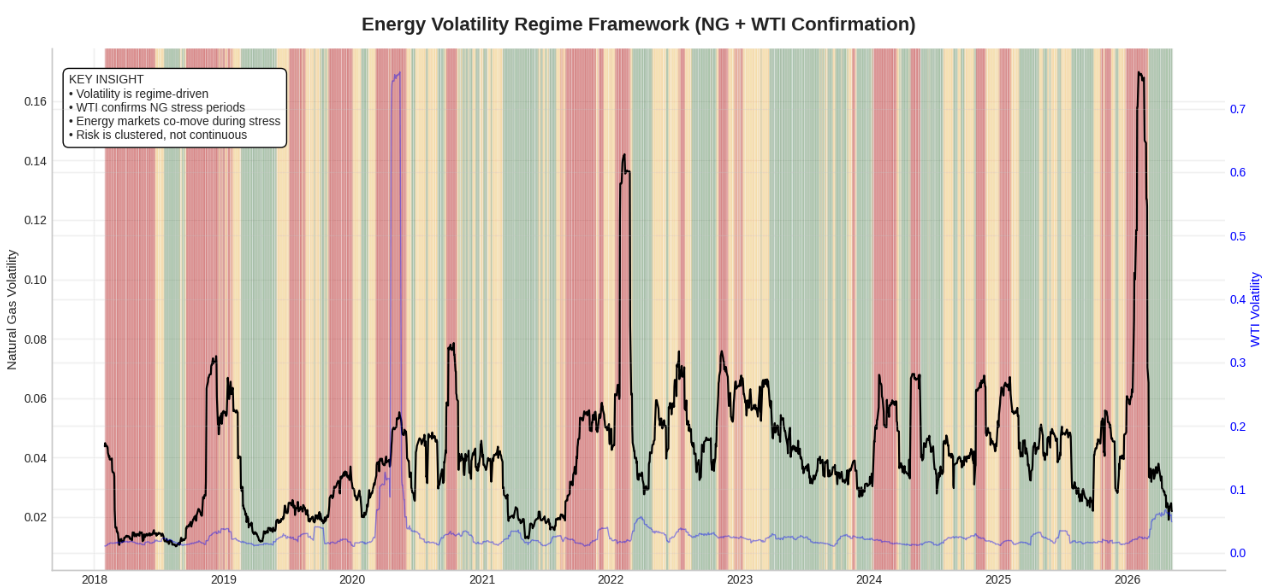


Figure Commentary

During calm market regimes, correlations between energy assets remain relatively dispersed, supporting diversification benefits.

However, during stress environments, cross-asset correlations increase materially, suggesting systemic convergence and reduced diversification effectiveness.

This behavior is consistent with institutional observations across broader commodity and macro markets during periods of elevated uncertainty.



Expected Return Analysis

The framework also evaluated expected returns conditional on volatility regime classification.

Results indicate that return distributions differ materially depending on the underlying market state, suggesting that volatility regime identification may improve positioning and risk management decisions.

Stress environments exhibited materially different return and volatility characteristics relative to calm market conditions.

Conclusion

This project demonstrates how volatility regime analysis can be applied to cross-commodity energy markets to better understand systemic stress propagation, diversification breakdowns, and market-state behavior.

The framework combines volatility analysis, spread monitoring, shock clustering, and correlation modeling into a unified institutional-style market risk structure intended to resemble components of real-world energy trading and risk analysis workflows.

Future enhancements could include:

- Regional basis market integration
- Weather sensitivity modeling
- Storage analytics
- Forward curve structure analysis
- Regime transition probability modeling
- Machine learning classification techniques

The broader objective of the project was to build a market-focused analytical framework emphasizing risk structure, volatility behavior, and cross-asset energy market dynamics rather than purely directional forecasting.

Appendix — Technical Framework

Programming Environment

- Python
- Google Colab
- pandas
- numpy
- matplotlib
- yfinance

Core Analytical Components

- Rolling volatility estimation
- Volatility regime classification
- Cross-commodity spread analysis
- Shock clustering detection
- Correlation regime analysis
- Expected return decomposition